

Supplementary Material

DACT: Dynamics-Aware Learning of Coupled Torque Policies for Disturbance-Robust Legged Locomotion

1 Training

1.1 Training Reward Structure

Reward	Equation (r_i)	Weight S1 (w_i^1)	Weight S2 (w_i^2)
Lin. velocity tracking	$\exp\{-4(\dot{\mathbf{p}}_{xy}^{ex} - \dot{\mathbf{p}}_{xy})^2\}$	1.0	1.0
Ang. velocity tracking	$\exp\{-4(\dot{\Theta}_{yaw}^{ex} - \dot{\Theta}_{yaw})^2\}$	0.5	0.5
DOF tracking reward	$\exp(-4\ \mathbf{q}^{des} - \mathbf{q}\ ^2)$	0.1	0.1
Coupled Torque Alignment	$\max(0, \cos_sim(\mathbf{a}^\tau, \tau^{fb}))$	0.40	0.40
Weighted Coupled Torque Conflict	$\frac{m_r}{m_r + \Delta m} \min(0, \cos_sim(\mathbf{a}^\tau, \tau^{fb}))$	0.10	0.10
Feedback torque magnitude	$\ \tau^{PD}\ ^2$	-2.0×10^{-5}	-2.0×10^{-5}
Feedforward torque magnitude	$\ \mathbf{a}^\tau\ ^2$	-1.0×10^{-5}	-1.0×10^{-5}
Linear velocity (z)	$\dot{\mathbf{p}}_z^2$	-2.0	-2.0
Angular velocity (xy)	$ \omega_{xy} ^2$	-0.05	-0.20
Orientations	$ \mathbf{g} ^2$	-0.2	-2.0
Joint accelerations	$\ \ddot{\mathbf{q}}\ ^2$	-2.5×10^{-7}	-2.5×10^{-7}
Joint power	$ \tau^{ex} \dot{\mathbf{q}} $	-2×10^{-5}	-2×10^{-5}
Body height	$(\mathbf{p}_z^{des} - \mathbf{p}_z)^2$	-2.0	-2.0
Action rate	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ ^2$	-0.001	-0.01
Smoothness	$(\mathbf{a}_t - 2\mathbf{a}_{t-1} + \mathbf{a}_{t-2})^2$	-0.001	-0.01
Power distribution	$\text{var}(\tau^{ex} \cdot \dot{\mathbf{q}})^2$	-10^{-5}	-10^{-5}
Joint torque limit violation	$\max(0, \tau^{ex} - \tau^{\max})^2 + \max(0, \tau^{\min} - \tau^{ex})^2$	-0.01	-1.0
Joint pose limit violation	$\max(0, \mathbf{q} - \mathbf{q}^{\max})^2 + \max(0, \mathbf{q}^{\min} - \mathbf{q})^2$	-2.0	-2.0
Feet contact force penalty	$\sum_k \max(0, \ \mathbf{f}^r\ - 200.0)$	-0.01	-0.01
Foot clearance	$(\mathbf{p}_{f,z,k}^{des} - \mathbf{p}_{f,z,k})^2 \cdot \mathbf{v}_{f,xy,k}$	-0.01	-0.01
Feet air-time reward	$\sum_k \mathbb{I}_{\text{first}}(0.5 - t_0) \mathbb{I}(\ \mathbf{v}_{cmd}\ > v_0)$	0.5	0.5
Foot slip penalty	$\sum_k \mathbb{I}(\mathbf{f}_{z,k}^r > \mathbf{f}_0^r) \ \mathbf{v}_{f,xy,k}\ ^2$	-0.1	-0.1
VHIP angle	$\max\left(0, \left \cos^{-1}\left(\frac{p_z}{\ell_{vhip}}\right) - \theta_0\right \right)$	-0.1	-0.1
VHIP angular acc.	$\max\left(0, \left \frac{g}{\ell_{vhip}} \sin(\theta_{vhip}) - \ddot{\theta}_0\right \right)$	-0.001	-0.001
Support polygon	$\begin{cases} \exp\left(-\frac{\ \mathbf{p}_b^{xy} - \mathbf{p}_S\ - 0.6r_S\ _+^2}{\sigma}\right), & S \geq 3 \\ \exp\left(-\frac{[d(\mathbf{p}_b^{xy}, \mathbf{p}_i \mathbf{p}_j) - 0.3\ \mathbf{p}_i - \mathbf{p}_j\ _+^2]}{\sigma}\right), & S = 2 \\ 0, & S < 2 \end{cases}$	0.20	0.20
Front foot overreach	$s_m \sum_{i \in \mathcal{F}} \mathbb{I}[f_i^z > 5] [x_i^b - x_{\max}]_+^2$	-100.0	-100.0
Rear foot overreach	$\sum_{i \in \mathcal{R}} \mathbb{I}[f_i^z > 5] [x_i^b - x_{\text{rear}}^{\text{nom}} - \delta_{\text{rear}}]_+^2$	-10.0	-10.0

Table 1. Reward terms, equations, and associated weights for each training curriculum stage. $(\cdot)^{ex}$ denotes the expected commanded values. $\exp(\cdot)$ and $\text{var}(\cdot)$ denote the exponential and variance operators respectively. An additional small alive bonus of 0.001 is added at each timestep and a collision penalty of -10.0 is also added for every link that experiences contact forces outside of the feet and calf links. **S1** and **S2** denote the weight values used in the first and second stages of the training curriculum respectively. The weight values were annealed with a cosine schedule from the values in **S1** to those in **S2** over 500 training iterations.

Each **DACT**-style coupled policy is trained with the reward formulation in Table 1. Weights in **S1** correspond to Stage I, *Dynamics-Aware Policy Discovery*, and weights in **S2** correspond to Stage II, *Disturbance Adaptation*; S2 weights are introduced with a 500-iteration cosine schedule starting at iteration 6,000.

Most terms are adapted from [1], with additional rewards and penalties designed for coupled torque–position control. A small DOF tracking reward encourages meaningful position targets by penalizing

deviations between realized joint positions and the postures induced by the coupled actions. Separate magnitude penalties are applied to feedforward torques $\mathbf{a}^\tau$ and feedback torques τ^{PD} , allowing asymmetric regularization of the two control channels. Action-rate and smoothness penalties are applied to the combined action $\mathbf{a} = \mathbf{a}^q \cup \mathbf{a}^\tau$, while joint pose and torque-limit penalties enforce kinematic and actuator feasibility. We also penalize excessive ground reaction forces and foot slip during contact.

To improve torso stability, we add VHIP-inspired penalties based on the virtual pendulum between the robot CoM and contact-force-weighted CoP [2]. The pendulum angle $\theta = \cos^{-1}(|p_{\text{CoM},z}|/\|\mathbf{p}_{\text{CoM}} - \mathbf{p}_{\text{CoP}}\|)$ is penalized above $\theta_{\text{th}} = 0.1$ rad to discourage large roll/pitch deviations, and the angular acceleration $\ddot{\theta} = g \sin(\theta)/\|\mathbf{p}_{\text{CoM}} - \mathbf{p}_{\text{CoP}}\|$ is penalized above $\ddot{\theta}_{\text{th}} = 0.01$ to encourage early correction toward stable configurations. A support-polygon reward further encourages the base projection $\mathbf{p}_b^{xy}$ to remain near the stance support set $S = \{i : f_i^z > 5\text{N}\}$: for $|S| \geq 3$, it penalizes distance beyond 0.6 of the stance radius r_S ; for $|S| = 2$, it penalizes lateral distance from the stance-foot segment beyond 0.3 of the segment length; and for $|S| < 2$, the reward is zero. The support-polygon reward σ value is $\sigma = 0.01$.

Finally, we include foot-overreach penalties to prevent stance feet from contacting the ground far from their nominal base-frame locations, which reduces the legs' ability to generate torso-stabilizing torques under payloads. Let $[x]_+ = \max(0, x)$, let x_i^b denote the i th foot's base-frame x position, and identify stance feet by $f_i^z > 5\text{N}$. The front-foot penalty discourages stance front feet $\mathcal{F}$ from exceeding $x_{\text{max}} = 0.28$, with mass-dependent scaling $s_m = 1 - \frac{1}{2} \frac{m_r}{m_r + m_p}$ to relax the constraint without added payloads. The rear-foot penalty keeps stance rear feet $\mathcal{R}$ near $x_{\text{rear}}^{\text{nom}} = -0.20$ within margin $\delta_{\text{rear}} = 0.08$, penalizing both excessive forward and backward placement.

The **DW** and **TO** baselines use the reward formulations from [1] and [3], respectively. For fair comparison, baselines also include all compatible rewards from Table 1 with identical weights.

1.2 Domain Randomization Details

Parameter	Range S1	Range S2	Unit
Payload (Δm_{robot})	$[-1, 2]$	$[-1, 8]$	kg
Planar Disturbance Impulse Velocity ($\mathbf{F}_{\text{ext}}^{xy}$)	$[-0.5, 0.5]$	$[-1.5, 1.5]$	ms
Vertical Disturbance Impulse Velocity ($\mathbf{F}_{\text{ext}}^z$)	$[-0.1, 0]$	$[-0.5, 0.0]$	ms
Disturbance Impulse Angular Velocity ($\mathbf{M}_{\text{ext}}$)	$[-0.5, 0.5]$	$[-1.5, 1.5]$	rad/s
Linear disturbance Interval	$[5.0, 15.0]$	$[5.0, 15.0]$	s
Angular disturbance Interval	$[5.0, 15.0]$	$[5.0, 15.0]$	s
K_p factor (ΔK_p)	$[0.8, 1.2]$	$[0.8, 1.2]$	Nm/rad
K_d factor (ΔK_d)	$[0.8, 1.2]$	$[0.8, 1.2]$	Nms/rad
Motor Strength Multiplicative Factor	$[0.9, 1.1]$	$[0.9, 1.1]$	N
Center of mass shift x (ΔCoM_x)	$[-0.075, 0.075]$	$[-0.20, -0.20]$	m
Center of mass shift y (ΔCoM_y)	$[-0.075, 0.075]$	$[-0.15, -0.15]$	m
Center of mass shift z (ΔCoM_z)	$[-0.075, 0.075]$	$[0.15, 0.15]$	m
Friction coefficient	$[0.2, 1.25]$	$[0.2, 1.25]$	—
Joint armature	$[0.00, 0.015]$	$[0.00, 0.015]$	Nms/rad
Joint friction	$[0.00, 0.05]$	$[0.00, 0.20]$	Nms/rad
Joint stiffness factor	$[0.00, 0.005]$	$[0.00, 0.020]$	—
Joint damping factor	$[0.20, 0.60]$	$[0.00, 0.80]$	—

Table 2. Domain randomization parameters and ranges. **S1** and **S2** denote the ranges used in the first and second stages of the training curriculum respectively.

The domain randomization ranges shared across all approaches are presented in Table 2. We follow standard randomized parameters from prior legged-locomotion work [1, 4, 3], and additionally randomize vertical disturbance impulses and 3-DOF torso moments. Torso disturbances are applied as impulse wrenches: the corresponding linear or angular impulse velocity is sampled from the current range and applied at randomized temporal intervals. The fairly large values for joint-level-dynamics randomization was found empirically to be critical for successful sim2real deployment of the trained **DACT** policy.

Algorithm 1: Performance-Gated Domain Randomization Curriculum

Input : Training iteration k ; mean tracking reward $\bar{r}_k$; progress variables $p_{\text{joint}}, p_{\text{mc}}, p_{\text{dist}} \in [0, 1]$; EMA factor α ; recovery ratio η ; minimum reward r_{min} ; step interval K_{step} ; progress increments $(\Delta_{\text{joint}}, \Delta_{\text{mc}}, \Delta_{\text{dist}})$

Output : Updated domain randomization ranges

```
1 Update reward EMA:  $\tilde{r}_k \leftarrow (1 - \alpha)\tilde{r}_{k-1} + \alpha\bar{r}_k$ 
2 Compute recent reference performance:  $r_k^* \leftarrow \text{Quantile}_{0.90}(\{\tilde{r}_i\}_{i=k-W}^k)$ 
3 Set stepping condition:  $\text{step} \leftarrow (\tilde{r}_k \geq \eta r_k^*) \wedge (\tilde{r}_k \geq r_{\text{min}}) \wedge (k - k_{\text{last}} \geq K_{\text{step}})$ 
4 if  $k \leq K_{\text{warmup}}$  or not step then
5   | Hold current randomization ranges
6 end
7 else
8   if phase is joint dynamics then
9     |  $p_{\text{joint}} \leftarrow \min(1, p_{\text{joint}} + \Delta_{\text{joint}})$ 
10  end
11  else if phase is mass/CoM then
12    |  $p_{\text{mc}} \leftarrow \min(1, p_{\text{mc}} + \Delta_{\text{mc}})$ 
13  end
14  else if phase is disturbance then
15    |  $p_{\text{dist}} \leftarrow \min(1, p_{\text{dist}} + \Delta_{\text{dist}})$ 
16  end
17  Advance to the next phase when the active progress reaches 1.0
18   $k_{\text{last}} \leftarrow k$ 
19 end
20 Update each randomized range by linear interpolation:  $\mathcal{R}_j \leftarrow \mathcal{R}_j^0 + p_{\text{joint}}(\mathcal{R}_j^1 - \mathcal{R}_j^0)$ 
21  $\mathcal{R}_{\text{mc}} \leftarrow \mathcal{R}_{\text{mc}}^0 + p_{\text{mc}}(\mathcal{R}_{\text{mc}}^1 - \mathcal{R}_{\text{mc}}^0)$ 
22  $\mathcal{R}_d \leftarrow \mathcal{R}_d^0 + p_{\text{dist}}(\mathcal{R}_d^1 - \mathcal{R}_d^0)$ 
```

We update randomization ranges using the performance-gated curriculum in Algorithm 1. After a warm-up period of $K_{\text{warmup}} = 2000$ iterations (corresponding to the completion of training stage **S1**) have passed, the curriculum progresses through three phases: joint-level dynamics randomization, payload/CoM randomization, and external disturbance randomization. At each iteration, an exponential moving average of the tracking reward is compared against both an absolute reward threshold and a recovery threshold defined by the 0.90 quantile of recent reward history. The randomization range advances only when performance has sufficiently recovered and at least $K_{\text{step}} = 10$ iterations have elapsed since the previous update. When these conditions are met, the active phase progress is increased by $\Delta_{\text{joint}} = 0.02$, $\Delta_{\text{mc}} = 0.01$, or $\Delta_{\text{dist}} = 0.01$, and the corresponding parameter ranges are linearly interpolated between their **S1** and **S2** bounds. This gating prevents the policy from being exposed to stronger dynamics perturbations before recovering stable tracking performance under the current randomization range. Because the threshold is based on an EMA relative to recent performance, the curriculum can still progress as tracking naturally degrades under harder disturbances. The same curriculum is applied uniformly to all approaches.

1.3 Algorithm for Constructing PINN Loss

The procedure used to construct the PINN losses introduced in the main text is summarized in Alg. 2. To compute the dynamics terms $\mathbf{A}$, $\mathbf{c}$, $\mathbf{g}$, and $\mathbf{J}_c$, we use the Pinocchio rigid-body dynamics library [5]. Linear and angular accelerations of the robot torso and the angular accelerations of the leg joints are approximated using a first-order backward finite difference of simulator-provided velocities. The Pinocchio robot model is then updated to account for the randomized torso CoM and payload mass. The generalized positions and velocities are passed to Pinocchio to compute whole-body dynamics terms. Specifically, Pinocchio’s recursive Newton–Euler algorithm (RNEA) is used to evaluate the generalized mass matrix $\mathbf{A}$, while the composite rigid body algorithm (CRBA) computes the combined Coriolis and gravity vector $(\mathbf{b}, \mathbf{g})$. Foot contact Jacobians are computed for each contacting foot and stacked to form the block contact Jacobian $\mathbf{J}_c$. All of the computed values are then used to calculate the whole-body dynamics loss. Since Pinocchio does not natively

Algorithm 2: Construct PINN Loss

Input: Linear velocities $\dot{\mathbf{p}}_t, \dot{\mathbf{p}}_{t+1}$; angular velocities $\dot{\boldsymbol{\Theta}}_t, \dot{\boldsymbol{\Theta}}_{t+1}$; joint velocities $\dot{\mathbf{q}}_t, \dot{\mathbf{q}}_{t+1}$; generalized coordinates $\boldsymbol{\rho}$ and velocities $\dot{\boldsymbol{\rho}}$; contact forces $\mathbf{f}_{t+1}^{r,c}$; coupled torques $\boldsymbol{\tau}_t^{ex}$; CoM shift ΔCoM ; Added Mass Δm ; time step Δt ; foot frame set $\mathcal{F}_{\text{feet}}$, Instantiated Pinocchio Robot Model $\mathcal{M}_{\text{pin}}$;

Output: PINN loss $\mathcal{L}_{\text{PINN}}$

- 1 Finite-difference base accelerations: $\ddot{\mathbf{p}}_{t+1} \leftarrow \frac{\dot{\mathbf{p}}_{t+1} - \dot{\mathbf{p}}_t}{\Delta t}$, $\ddot{\boldsymbol{\Theta}}_{t+1} \leftarrow \frac{\dot{\boldsymbol{\Theta}}_{t+1} - \dot{\boldsymbol{\Theta}}_t}{\Delta t}$, $\ddot{\mathbf{q}}_{t+1} \leftarrow \frac{\dot{\mathbf{q}}_{t+1} - \dot{\mathbf{q}}_t}{\Delta t}$
 - 2 Assemble generalized acceleration vector $\ddot{\boldsymbol{\rho}}_t = [\ddot{\mathbf{p}}_{t+1}, \ddot{\boldsymbol{\Theta}}_{t+1}, \ddot{\mathbf{q}}_t]^\top$
 - 3 Update Pinocchio inertial model with randomized torso parameters:
 $\mathcal{M}_{\text{pin}} \leftarrow \text{updateInertialParams}(\mathcal{M}_{\text{pin}}, \Delta m, \Delta\text{CoM})$
 - 4 Compute generalized mass matrix term $\mathbf{A} \leftarrow \text{pinocchio.RNEA}(\mathcal{M}_{\text{pin}}, \boldsymbol{\rho}_{t+1}, \dot{\boldsymbol{\rho}}_{t+1}, \mathbf{0})$
 - 5 Compute Coriolis and gravity terms: $\mathbf{b}, \mathbf{g} \leftarrow \text{pinocchio.CRBA}(\mathcal{M}_{\text{pin}}, \boldsymbol{\rho}_{t+1})$
 - 6 Initialize contact Jacobian list: $\mathcal{J} \leftarrow \emptyset$
 - 7 **foreach** foot frame $f \in \mathcal{F}_{\text{feet}}$ **do**
 - 8 Retrieve foot frame index: $i_f \leftarrow \text{getFrameId}(f)$
 - 9 Compute foot Jacobian: $\mathbf{J}_f \leftarrow \text{pinocchio.computeFrameJacobian}(\boldsymbol{\rho}, i_f)$
 - 10 Append to contact Jacobian set: $\mathcal{J} \leftarrow \mathcal{J} \cup \mathbf{J}_f$
 - 11 Stack contact Jacobians: $\mathbf{J}_c \leftarrow \text{concat}(\mathcal{J}, \text{axis} = 0)$
 - 12 Compute PINN residual loss: $\mathcal{L}_{\text{PINN}} \leftarrow \left\| \mathbf{A}\ddot{\boldsymbol{\rho}}_{t+1} + \mathbf{c}\dot{\boldsymbol{\rho}}_{t+1} + \mathbf{g} - \begin{pmatrix} \mathbf{0}_6 \\ \boldsymbol{\tau}_t^{ex} \end{pmatrix} - \mathbf{J}_c^\top \mathbf{f}_{t+1}^{r,c} \right\|^2$
 - 13 **return** $\mathcal{L}_{\text{PINN}}$
-

support batch computation, the steps in Alg. 2 are executed independently for each sample in the batch. To mitigate the resulting computational cost, we parallelize these computations as batch operations using Python multiprocessing with shared and persistent memory across CPU cores. The resulting dynamics quantities are then reshaped into batch form and used to evaluate the PINN loss in a vectorized manner. In practice, the dynamics quantities $\mathbf{A}$, $\mathbf{b}$, $\mathbf{g}$, and $\mathbf{J}_c$ are computed during the PPO simulation step and cached for subsequent loss evaluation. They are presented inline with the loss computation in Alg. 2 for clarity of exposition.

1.4 Training Algorithm

Algorithm 3 summarizes the training procedure for **DACT**, which combines on-policy reinforcement learning with privileged-dynamics reconstruction and physics-informed action regularization. During rollout, the policy maintains an observation history that is encoded into a latent context $\mathbf{z}_t$ and base-velocity estimate $\hat{\mathbf{p}}_t$. Conditioned on this context and the current observation, the coupled policy outputs joint-position actions $\mathbf{a}_t^q$ for feedback-stabilized gait regulation and feedforward torques $\mathbf{a}_t^r$ for disturbance compensation. The resulting transitions, privileged observations, and policy log-likelihoods are stored in an on-policy buffer, after which returns and advantages are computed using the standard PPO procedure.

Policy optimization uses a composite objective that combines the clipped PPO loss, privileged reconstruction loss, and PINN regularization. The reconstruction loss trains the latent context to predict the next-step privileged observation $\mathbf{o}_{t+1}^p$, encouraging $\mathbf{z}_t$ to encode disturbance-relevant dynamics such as payload variation, CoM shifts, external torso wrenches, randomized gains, and contact forces. The PINN loss, constructed in Algorithm 2, penalizes violations of the disturbed whole-body dynamics induced by the coupled torque command, thereby biasing the learned action representation toward physically consistent stabilization. The encoder, policy, and value function are updated jointly, while the decoder is updated in a separate step using a frozen encoder. In practice, we additionally optimize only the encoder using the KL-divergence and reconstruction objectives using another feedforward pass without subsequently invoking the actor network, but we exclude this from Alg. 3 for the sake of compactness. Any potential gradient conflicts between $\mathcal{L}_{\text{PPO}}$ and $\mathcal{L}_{\text{PINN}}$ are resolved using the prioritized PCGrad [6] gradient projection technique introduced in [7],

Algorithm 3: DACT PPO

Input : Encoder $\Upsilon_{\psi}^{\text{ENC}}$, decoder $\Upsilon_{\delta}^{\text{DEC}}$, coupled policy π_{θ}^c , value function V_{ϕ} ; environment $\mathcal{E}$; rollout horizon T ; discount γ ; PPO clip ϵ ; value and entropy weights (c_v, c_e)
Output : Updated parameters $(\psi, \delta, \theta, \phi)$

```
1 Initialize networks and on-policy buffer  $\mathcal{D}$ 
2 for iteration  $k = 1, 2, \dots$  do
3   Rollout phase
4   for  $t = 1$  to  $T$  do
5     Observe  $\mathbf{o}_t$  and privileged observation  $\mathbf{o}_t^p$ 
6     Update history  $\mathbf{o}_t^H \leftarrow \mathbf{o}_{t-1}^H[1:] \cup \mathbf{o}_t$ 
7     Encode privileged-dynamics-aware context  $(\mathbf{z}_t, \hat{\mathbf{p}}_t) \sim \Upsilon_{\psi}^{\text{ENC}}(\mathbf{o}_t^H)$ 
8     Sample coupled action  $(\mathbf{a}_t^q, \mathbf{a}_t^r) \sim \pi_{\theta}^c(\cdot \mid \mathbf{o}_t, \mathbf{z}_t, \hat{\mathbf{p}}_t)$ 
9     Execute  $(\mathbf{a}_t^q, \mathbf{a}_t^r)$  in  $\mathcal{E}$  and observe  $(r_t, \mathbf{o}_{t+1}, \mathbf{o}_{t+1}^p, d_t)$ 
10    Store transition, privileged quantities, and log probability in  $\mathcal{D}$ 
11  end
12  Compute bootstrapped returns and normalized advantages  $\hat{R}_t, \hat{A}_t$ 
13  Policy optimization phase
14  foreach minibatch  $\mathcal{B} \subset \mathcal{D}$  do
15    Sample  $(\mathbf{z}_t, \hat{\mathbf{p}}_t) \sim \Upsilon_{\psi}^{\text{ENC}}(\mathbf{o}_t^H)$ 
16    Compute clipped PPO objective  $\mathcal{L}_{\text{PPO}}$  using  $\pi_{\theta}^c$  and  $V_{\phi}$ 
17    Reconstruct next-step privileged observation using frozen decoder:  $\hat{\mathbf{o}}_{t+1}^p \leftarrow \Upsilon_{\delta}^{\text{DEC}}(\mathbf{z}_t, \hat{\mathbf{p}}_t)$ 
18    Compute privileged reconstruction loss:  $\mathcal{L}_{\text{dyn}} \leftarrow \|\hat{\mathbf{o}}_{t+1}^p - \mathbf{o}_{t+1}^p\|_2^2 + \mathcal{L}_{\text{KD}}$ 
19    Compute physics-informed action regularization:  $\mathcal{L}_{\text{PINN}} \leftarrow \text{ConstructPINNLoss}(\mathcal{B})$  using
      Algorithm 2
20    Compute total loss:  $\mathcal{L}_{\text{total}} \leftarrow \text{PPCGrad}(\mathcal{L}_{\text{PPO}}, \mathcal{L}_{\text{PINN}}) + \mathcal{L}_{\text{dyn}}$ 
21    Update  $(\psi, \theta, \phi)$  using gradient descent on  $\mathcal{L}_{\text{total}}$ 
22    Decoder optimization
23    Sample  $(\mathbf{z}_t, \hat{\mathbf{p}}_t)$  from frozen  $\Upsilon_{\psi}^{\text{ENC}}$ 
24    Reconstruct  $\hat{\mathbf{o}}_{t+1}^p \leftarrow \Upsilon_{\delta}^{\text{DEC}}(\mathbf{z}_t, \hat{\mathbf{p}}_t)$ 
25    Update  $\delta$  using  $\mathcal{L}_{\text{DEC}} = \|\hat{\mathbf{o}}_{t+1}^p - \mathbf{o}_{t+1}^p\|_2^2$ 
26  end
27  Apply layerwise spectral normalization to the encoder and actor:  $\Upsilon_{\psi}^{\text{ENC}} \leftarrow \text{SpecNorm}(\Upsilon_{\psi}^{\text{ENC}})$ ,
     $\pi_{\theta}^c \leftarrow \text{SpecNorm}(\pi_{\theta}^c)$ 
28 end
```

where $\mathcal{L}_{\text{PPO}}$ is the prioritized gradient. Finally, we apply layer-wise spectral normalization [8, 9] to the encoder and actor networks, with a maximum layer spectral norm of 6.0. Layer-wise spectral normalization rescales each linear layer by its largest singular value, bounding the layer’s Lipschitz constant and limiting how strongly small input changes can amplify through the network. For **DACT**, this stabilizes the encoder and coupled actor by reducing abrupt changes in the latent context, feedforward torques, and position targets, which helps mitigate overly aggressive torque responses during disturbance-rich training and physical deployment.

In practice, we found that training **DACT** policies from scratch with Algorithm 3 can occasionally converge to asymmetric gait patterns. In these cases, one lateral pair of legs learns a “reaching” gait that steps beyond the robot’s current body position, while the opposite pair learns a compensatory “matching” gait that advances only far enough to align with the reaching feet. We eliminate this failure mode with a simple torque-matching pretraining stage: for 2,000 iterations, only the position-control outputs are used to control the robot, while the feedforward torque head is trained to imitate the PD torques induced by those position targets. This initializes the coupled action space with a dynamically consistent relationship between the position and torque channels before full **DACT** training. During torque-matching pretraining, we use the Stage I reward weights and domain-randomization ranges, with a reduced torque-limit penalty weight of 0.001 and reduced torso-height and vertical-velocity penalties of 1.0 each; all other hyperparameters are unchanged. For fairness, all ablations and

baselines are trained for the same 2,000-iteration pretraining phase under these simplified settings before beginning Stage I.

1.5 Hyperparameters

Table 3. **DACT** Hyperparameters

Hyperparameter	Value
Num. Iterations S1	2000
Num. Iterations S2	6000
Num. Environments	4096
Steps Per Iteration	32
Epochs Per Iteration	5
Num. Mini-Batches	4
π^c , Υ_{enc} , and Υ_{dec} Learning Rate	3×10^{-4}
Critic Learning Rate	1×10^{-4}
Discount Factor γ	0.99
GAE λ	0.95
PPO Clip	0.2
Gradient Norm Clip	1.0
Entropy Coefficient Range	[0.005, 0.01]
Max Episode Length	20s
Action Function	elu
Encoder Υ_{enc} Layers	[128, 64, 19]
Decoder Υ_{dec} Layers	[64, 128]
Value Network Layers	[1024, 256, 128]
Policy π^c Layers	[512, 256, 128]
Policy Frequency	100Hz
PD-Feedback Frequency	500Hz

Algorithm 4: Adaptive Entropy Coefficient

Input : Performance metrics: linear tracking r_{lin} , angular tracking r_{ang} , terrain progress ℓ_{ter} ; thresholds $(\bar{r}_{\text{lin}}, \bar{r}_{\text{ang}}, \bar{\ell}_{\text{ter}})$; entropy bounds $[\beta_{\text{min}}, \beta_{\text{max}}]$; softmax temperature T_{ent}

Output : Entropy coefficient β

- 1 Compute normalized performance gaps:

$$g_{\text{lin}} = \frac{[\bar{r}_{\text{lin}} - r_{\text{lin}}]_+}{\bar{r}_{\text{lin}}}, \quad g_{\text{ang}} = \frac{[\bar{r}_{\text{ang}} - r_{\text{ang}}]_+}{\bar{r}_{\text{ang}}}, \quad g_{\text{ter}} = \frac{[\bar{\ell}_{\text{ter}} - \ell_{\text{ter}}]_+}{\bar{\ell}_{\text{ter}}}$$

- 2 Compute softmax weights: $\mathbf{w} \leftarrow \text{softmax}([g_{\text{lin}}, g_{\text{ang}}, g_{\text{ter}}]/T_{\text{ent}})$
 - 3 Compute weighted gap: $g \leftarrow \mathbf{w}^\top [g_{\text{lin}}, g_{\text{ang}}, g_{\text{ter}}]$
 - 4 Update entropy coefficient: $\beta \leftarrow \beta_{\text{min}} + g(\beta_{\text{max}} - \beta_{\text{min}})$
 - 5 **return** β
-

Table 3 summarizes the training hyperparameters. Base velocity commands are sampled uniformly from $[-1.0, 1.0]$ across all DOFs. During early training, the linear velocity range is restricted to $[-0.5, 0.5]$ m/s and expanded once $\pi^c(\cdot)$ achieves an average episodic linear tracking reward of 0.8. The context encoder $\Upsilon_{\text{enc}}(\cdot)$ processes a history of $H = 20$ timesteps, effectively maintaining a history of 0.20 seconds. The training terrains consist of 10×20 subterrains of size 8×8 m and a 4 m central platform. Terrain types are sampled as smooth slopes, rough slopes, up/down stairs, discrete obstacles, and wave terrain with proportions $[0.10, 0.15, 0.25, 0.25, 0.20, 0.05]$. Difficulty scales each terrain level, increasing slopes up to 0.4, roughness from 0.02–0.12 m, stair and obstacle heights from 0.04–0.20 m, and wave amplitude up to 0.20 m.

Additionally, for all approaches, we use the adaptive entropy schedule in Algorithm 4 to maintain exploration when tracking or terrain progression degrades, as proposed in [10]. The entropy coefficient is initialized to 0.01 and constrained to $[0.005, 0.01]$. At each update, we compute normalized gaps between the current performance metrics and target thresholds for linear velocity tracking (0.75), angular velocity tracking (0.35), and average terrain curriculum level (6.0). A temperature-scaled

softmax with $T_{\text{ent}} = 2.0$ weights these gaps, producing a scalar performance deficit used to interpolate the entropy coefficient between its lower and upper bounds. Thus, entropy decreases as the policy achieves stable tracking and terrain progress, but increases when performance falls below the specified targets.

2 Additional Experimental Results

2.1 PINN Loss Training Performance

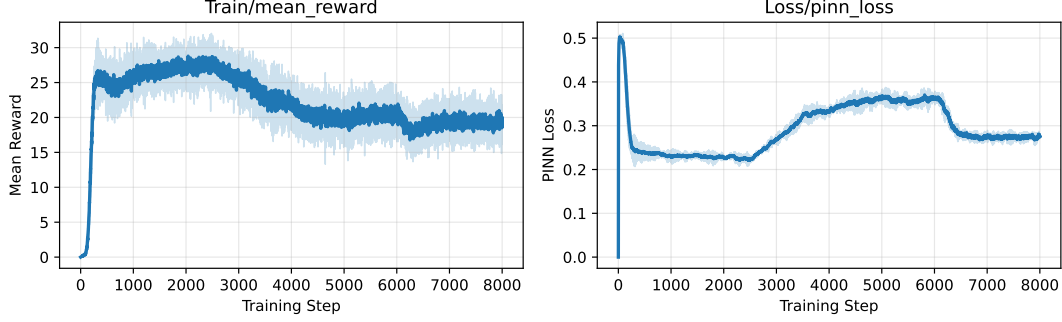


Figure 1. **DACT** mean episodic reward and $\mathcal{L}_{\text{PINN}}$ loss performance averaged (with two standard deviations) across three random seeds.

Figure 1 shows that **DACT** rapidly learns a stable locomotion policy, with mean episodic reward increasing sharply early in training and remaining high across three random seeds. The reward gradually decreases after the curriculum becomes more difficult, reflecting the transition to stronger disturbances and broader domain randomization, but stabilizes by the end of training. The PINN loss initially spikes, then settles before rising during the more challenging curriculum phases, indicating that the physics regularizer becomes more active as the policy encounters harder dynamics. Its later decrease and stabilization suggest that **DACT** adapts to these disturbances while maintaining improved whole-body dynamics consistency.

2.2 Simulated Evaluation Per-Terrain Results

Method	# Fail.↓	$\mathbf{v}_{x,y}^e$ ↓	ω_{yaw}^e ↓	$\mathbf{q}_j^e$ ↓	p_z^e ↓	$\ \mathbf{g}\ _1$ ↓	v_z ↓	$\ \omega\ _1$ ↓	$\ \tau\dot{\mathbf{q}}\ _1$ ↓	P.A.↑
No Disturbance – Plane Terrain										
DW	0	0.102	0.133	0.164	0.049	0.049	0.050	0.532	40.039	–
TO	0	0.115	0.113	0.290	0.009	0.069	0.057	0.554	43.706	–
A1	0	0.107	0.108	0.084	0.052	0.055	0.052	0.513	36.586	-0.710
A2	0	0.106	0.114	0.097	0.043	0.058	0.044	0.523	39.713	0.795
A3	1	0.112	0.126	0.103	0.034	0.087	0.052	0.526	35.529	0.818
DC	0	0.107	0.111	0.094	0.043	0.056	0.049	0.481	38.107	0.783
RA	0	0.135	0.140	0.141	0.055	0.046	0.062	0.511	39.029	0.116
Payload and CoM Shifts – Plane Terrain										
DW	77	0.162	0.204	0.204	0.037	0.161	0.088	0.784	46.329	–
TO	108	0.169	0.173	0.384	0.033	0.164	0.094	0.798	54.720	–
A1	49	0.144	0.162	0.107	0.037	0.114	0.074	0.712	42.582	-0.503
A2	49	0.146	0.171	0.118	0.031	0.110	0.073	0.710	44.199	0.784
A3	41	0.148	0.167	0.122	0.031	0.123	0.073	0.668	40.784	0.806
DC	28	0.144	0.151	0.114	0.034	0.094	0.067	0.606	43.790	0.775
RA	52	0.193	0.229	0.161	0.044	0.174	0.113	1.017	56.303	0.381
External 6-DOF Wrenches – Plane Terrain										
DW	65	0.312	0.238	0.185	0.047	0.105	0.122	0.980	63.611	–
TO	47	0.325	0.205	0.305	0.014	0.123	0.129	0.918	73.983	–
A1	16	0.295	0.197	0.099	0.050	0.092	0.103	0.865	58.230	-0.570
A2	19	0.298	0.202	0.110	0.042	0.096	0.095	0.883	60.127	0.787
A3	30	0.296	0.210	0.114	0.035	0.117	0.104	0.851	54.719	0.807
DC	7	0.291	0.186	0.105	0.043	0.086	0.090	0.804	57.017	0.770
RA	124	0.327	0.264	0.169	0.051	0.118	0.141	1.051	65.731	0.182

Table 4. Plane terrain performance under each disturbance condition. Rows are grouped by terrain and disturbance setting; columns use the same metric definitions as in the main text. Metrics reporting norms use the L1 norm and all other metrics report MAE.

Method	# Fail.↓	$\mathbf{v}_{x,y}^e$ ↓	ω_{yaw}^e ↓	$\mathbf{q}_j^e$ ↓	p_z^e ↓	$\ \mathbf{g}\ _1$ ↓	v_z ↓	$\ \omega\ _1$ ↓	$\ \tau\dot{\mathbf{q}}\ _1$ ↓	P.A.↑
No Disturbance – Slope Terrain										
DW	0	0.129	0.150	0.170	0.382	0.108	0.072	0.637	43.567	–
TO	1	0.132	0.122	0.302	0.425	0.093	0.080	0.595	45.020	–
A1	0	0.132	0.135	0.093	0.367	0.119	0.073	0.627	42.523	-0.684
A2	0	0.130	0.135	0.102	0.375	0.121	0.065	0.623	43.853	0.774
A3	0	0.132	0.145	0.107	0.386	0.134	0.069	0.625	39.244	0.801
DC	0	0.132	0.134	0.097	0.374	0.115	0.067	0.597	41.218	0.756
RA	0	0.165	0.156	0.153	0.364	0.122	0.082	0.641	45.640	0.108
Payload and CoM Shifts – Slope Terrain										
DW	84	0.185	0.223	0.204	0.419	0.190	0.109	0.864	51.075	–
TO	115	0.194	0.180	0.384	0.460	0.177	0.111	0.830	55.700	–
A1	61	0.181	0.199	0.117	0.412	0.172	0.099	0.819	50.514	-0.474
A2	66	0.181	0.188	0.122	0.419	0.163	0.094	0.817	50.887	0.764
A3	40	0.178	0.189	0.124	0.424	0.171	0.090	0.754	45.387	0.789
DC	25	0.170	0.171	0.112	0.407	0.138	0.083	0.679	47.521	0.746
RA	98	0.218	0.254	0.172	0.423	0.217	0.136	1.099	62.749	0.273
External 6-DOF Wrenches – Slope Terrain										
DW	99	0.328	0.268	0.194	0.426	0.150	0.145	1.100	70.121	–
TO	84	0.336	0.222	0.318	0.448	0.142	0.157	0.977	78.056	–
A1	45	0.328	0.234	0.111	0.397	0.153	0.131	1.027	70.374	-0.530
A2	34	0.317	0.227	0.113	0.403	0.154	0.117	0.977	66.846	0.767
A3	34	0.316	0.239	0.119	0.417	0.167	0.125	0.967	61.855	0.792
DC	16	0.319	0.218	0.109	0.402	0.142	0.114	0.924	63.302	0.751
RA	121	0.350	0.298	0.175	0.414	0.176	0.170	1.190	71.276	0.209

Table 5. Slope terrain performance under each disturbance condition. Rows are grouped by terrain and disturbance setting; columns use the same metric definitions as in the main text. Metrics reporting norms use the L1 norm and all other metrics report MAE.

Method	# Fail.↓	$\mathbf{v}_{x,y}^e$ ↓	ω_{yaw}^e ↓	$\mathbf{q}_j^e$ ↓	p_z^e ↓	$\ \mathbf{g}\ _1$ ↓	v_z ↓	$\ \omega\ _1$ ↓	$\ \tau\dot{\mathbf{q}}\ _1$ ↓	P.A.↑
No Disturbance – Rough Terrain										
DW	0	0.126	0.162	0.169	0.045	0.079	0.079	0.684	43.800	–
TO	0	0.131	0.130	0.298	0.018	0.083	0.077	0.624	46.394	–
A1	0	0.125	0.143	0.090	0.049	0.091	0.084	0.667	41.600	-0.678
A2	0	0.125	0.144	0.102	0.041	0.088	0.076	0.663	43.657	0.777
A3	0	0.129	0.152	0.106	0.032	0.107	0.075	0.652	39.658	0.805
DC	0	0.125	0.138	0.097	0.040	0.082	0.072	0.618	41.446	0.764
RA	0	0.152	0.173	0.149	0.052	0.086	0.092	0.678	43.412	0.137
Payload and CoM Shifts – Rough Terrain										
DW	107	0.183	0.243	0.202	0.038	0.180	0.115	0.907	53.441	–
TO	131	0.184	0.185	0.380	0.035	0.172	0.113	0.843	56.551	–
A1	67	0.165	0.199	0.113	0.037	0.141	0.104	0.863	50.329	-0.462
A2	80	0.168	0.204	0.120	0.035	0.138	0.102	0.847	50.320	0.768
A3	59	0.165	0.196	0.123	0.031	0.140	0.097	0.782	44.997	0.799
DC	45	0.161	0.179	0.113	0.034	0.113	0.088	0.708	47.349	0.750
RA	112	0.210	0.256	0.167	0.047	0.182	0.138	1.113	58.802	0.279
External 6-DOF Wrenches – Rough Terrain										
DW	114	0.318	0.290	0.192	0.044	0.139	0.156	1.162	72.669	–
TO	92	0.325	0.228	0.309	0.021	0.138	0.152	0.991	78.458	–
A1	24	0.316	0.244	0.108	0.047	0.129	0.140	1.059	70.298	-0.518
A2	28	0.307	0.244	0.114	0.039	0.126	0.129	1.035	67.982	0.767
A3	52	0.304	0.253	0.118	0.033	0.140	0.132	0.994	64.167	0.796
DC	14	0.304	0.229	0.110	0.039	0.113	0.118	0.958	65.378	0.753
RA	154	0.344	0.317	0.177	0.048	0.157	0.182	1.247	74.676	0.188

Table 6. Rough terrain performance under each disturbance condition. Rows are grouped by terrain and disturbance setting; columns use the same metric definitions as in the main text. Metrics reporting norms use the L1 norm and all other metrics report MAE.

Method	# Fail.↓	$\mathbf{v}_{x,y}^e$ ↓	ω_{yaw}^e ↓	$\mathbf{q}_j^e$ ↓	p_z^e ↓	$\ \mathbf{g}\ _1$ ↓	v_z ↓	$\ \omega\ _1$ ↓	$\ \tau\dot{\mathbf{q}}\ _1$ ↓	P.A.↑
No Disturbance – Stairs Terrain										
DW	3	0.142	0.180	0.175	0.355	0.119	0.094	0.738	47.859	–
TO	0	0.144	0.139	0.307	0.397	0.100	0.096	0.648	49.812	–
A1	0	0.142	0.162	0.097	0.344	0.129	0.094	0.706	46.752	-0.653
A2	1	0.141	0.156	0.104	0.346	0.130	0.083	0.695	46.877	0.768
A3	0	0.148	0.170	0.109	0.363	0.140	0.083	0.698	43.199	0.797
DC	2	0.146	0.159	0.101	0.359	0.121	0.082	0.665	44.811	0.752
RA	0	0.172	0.182	0.158	0.335	0.134	0.103	0.734	50.330	0.105
Payload and CoM Shifts – Stairs Terrain										
DW	106	0.198	0.250	0.211	0.405	0.190	0.125	0.951	56.018	–
TO	123	0.210	0.205	0.382	0.441	0.190	0.124	0.884	60.504	–
A1	74	0.192	0.225	0.120	0.391	0.178	0.116	0.904	57.927	-0.441
A2	77	0.190	0.208	0.123	0.393	0.172	0.109	0.868	54.619	0.758
A3	76	0.190	0.221	0.128	0.405	0.172	0.105	0.822	49.649	0.791
DC	31	0.185	0.196	0.117	0.392	0.146	0.099	0.753	52.497	0.748
RA	139	0.223	0.277	0.173	0.403	0.226	0.148	1.154	66.049	0.236
External 6-DOF Wrenches – Stairs Terrain										
DW	140	0.325	0.302	0.197	0.390	0.167	0.167	1.210	77.450	–
TO	132	0.338	0.245	0.321	0.430	0.150	0.172	1.036	86.997	–
A1	70	0.326	0.269	0.115	0.376	0.162	0.152	1.124	79.754	-0.504
A2	89	0.321	0.261	0.116	0.398	0.163	0.138	1.079	73.875	0.765
A3	78	0.312	0.273	0.121	0.402	0.171	0.140	1.057	68.345	0.785
DC	44	0.317	0.253	0.113	0.380	0.149	0.131	1.019	72.524	0.741
RA	178	0.352	0.340	0.184	0.396	0.190	0.193	1.322	81.690	0.229

Table 7. Stairs terrain performance under each disturbance condition. Rows are grouped by terrain and disturbance setting; columns use the same metric definitions as in the main text. Metrics reporting norms use the L1 norm and all other metrics report MAE.

Method	# Fail.↓	$\mathbf{v}_{x,y}^e$ ↓	ω_{yaw}^e ↓	$\mathbf{q}_j^e$ ↓	p_z^e ↓	$\ \mathbf{g}\ _1$ ↓	v_z ↓	$\ \omega\ _1$ ↓	$\ \tau\dot{\mathbf{q}}\ _1$ ↓	P.A.↑
No Disturbance – Discrete Terrain										
DW	0	0.124	0.157	0.169	0.050	0.074	0.068	0.655	44.167	–
TO	1	0.131	0.129	0.297	0.026	0.084	0.072	0.620	46.590	–
A1	3	0.127	0.140	0.091	0.053	0.085	0.072	0.637	42.524	-0.679
A2	3	0.124	0.138	0.100	0.048	0.084	0.063	0.631	43.100	0.779
A3	0	0.128	0.151	0.106	0.041	0.103	0.065	0.625	39.508	0.807
DC	1	0.126	0.139	0.097	0.047	0.079	0.065	0.593	41.684	0.762
RA	4	0.153	0.164	0.147	0.057	0.080	0.080	0.638	43.389	0.127
Payload and CoM Shifts – Discrete Terrain										
DW	108	0.173	0.229	0.204	0.042	0.163	0.104	0.875	51.468	–
TO	122	0.185	0.184	0.371	0.042	0.172	0.106	0.840	56.792	–
A1	74	0.168	0.199	0.114	0.043	0.142	0.094	0.827	50.218	-0.466
A2	75	0.164	0.186	0.119	0.040	0.132	0.088	0.801	49.434	0.773
A3	85	0.175	0.204	0.123	0.040	0.153	0.093	0.773	46.786	0.792
DC	42	0.163	0.179	0.113	0.043	0.113	0.082	0.685	49.439	0.754
RA	141	0.209	0.258	0.168	0.048	0.197	0.129	1.069	60.986	0.231
External 6-DOF Wrenches – Discrete Terrain										
DW	128	0.331	0.294	0.193	0.049	0.139	0.150	1.171	74.145	–
TO	101	0.334	0.229	0.311	0.028	0.138	0.151	0.997	80.717	–
A1	51	0.308	0.238	0.107	0.051	0.122	0.124	1.023	69.291	-0.532
A2	62	0.312	0.236	0.112	0.046	0.123	0.118	0.999	68.928	0.777
A3	59	0.301	0.249	0.117	0.041	0.139	0.119	0.966	62.135	0.797
DC	29	0.305	0.225	0.110	0.046	0.109	0.110	0.935	66.736	0.760
RA	187	0.344	0.313	0.178	0.054	0.153	0.171	1.233	74.800	0.202

Table 8. Discrete terrain performance under each disturbance condition. Rows are grouped by terrain and disturbance setting; columns use the same metric definitions as in the main text. Metrics reporting norms use the L1 norm and all other metrics report MAE.

Method	# Fail.↓	$\mathbf{v}_{x,y}^e$ ↓	ω_{yaw}^e ↓	$\mathbf{q}_j^e$ ↓	p_z^e ↓	$\ \mathbf{g}\ _1$ ↓	v_z ↓	$\ \omega\ _1$ ↓	$\ \tau\dot{\mathbf{q}}\ _1$ ↓	P.A.↑
No Disturbance – Wave Terrain										
DW	6	0.122	0.159	0.167	0.084	0.101	0.073	0.662	42.246	–
TO	6	0.129	0.130	0.296	0.074	0.093	0.079	0.612	45.910	–
A1	4	0.124	0.140	0.090	0.084	0.113	0.073	0.636	40.926	-0.679
A2	4	0.127	0.139	0.099	0.084	0.111	0.069	0.640	42.973	0.768
A3	4	0.130	0.152	0.105	0.077	0.125	0.074	0.645	39.047	0.795
DC	1	0.124	0.133	0.095	0.082	0.099	0.071	0.602	40.664	0.749
RA	3	0.149	0.164	0.148	0.069	0.113	0.085	0.646	43.761	0.140
Payload and CoM Shifts – Wave Terrain										
DW	104	0.176	0.226	0.197	0.081	0.171	0.109	0.859	47.492	–
TO	131	0.190	0.185	0.368	0.082	0.177	0.116	0.863	57.029	–
A1	86	0.169	0.191	0.112	0.082	0.155	0.103	0.809	50.931	-0.490
A2	91	0.166	0.188	0.114	0.080	0.155	0.097	0.787	48.977	0.752
A3	71	0.171	0.191	0.122	0.078	0.156	0.098	0.766	44.358	0.789
DC	41	0.166	0.178	0.112	0.078	0.129	0.090	0.698	49.132	0.748
RA	101	0.203	0.244	0.161	0.057	0.191	0.129	1.042	55.949	0.331
External 6-DOF Wrenches – Wave Terrain										
DW	93	0.331	0.274	0.190	0.086	0.149	0.147	1.117	70.170	–
TO	91	0.337	0.230	0.313	0.078	0.142	0.157	1.003	81.058	–
A1	50	0.329	0.243	0.110	0.084	0.147	0.133	1.049	70.132	-0.531
A2	63	0.307	0.239	0.113	0.081	0.145	0.124	1.019	66.779	0.763
A3	41	0.302	0.242	0.116	0.081	0.153	0.124	0.961	60.097	0.790
DC	19	0.301	0.220	0.108	0.079	0.123	0.115	0.933	63.200	0.748
RA	146	0.348	0.304	0.174	0.090	0.171	0.176	1.222	71.389	0.216

Table 9. Wave terrain performance under each disturbance condition. Rows are grouped by terrain and disturbance setting; columns use the same metric definitions as in the main text. Metrics reporting norms use the L1 norm and all other metrics report MAE.

3 Limitations



Figure 2. **DACT** failure case when carrying 6.5 L liquid payload while walking up a stair case with 12 cm high steps.

Figure 2 shows a failure case in which **DACT** attempts to climb 12 cm stairs while carrying a 6.5 L liquid payload. During ascent, the front-right foot contacts and becomes caught on the inward bevel of the vertical stair face. The policy attempts to pull the foot over the stair edge, but the resulting torque demand triggers the Go1's built-in safety cutoff, disabling the motors. We attribute this failure to two limitations of the current formulation: **DACT** is a blind policy and therefore cannot proactively adjust swing-foot placement around sharp terrain discontinuities, and its relatively stiff torque response can produce excessive corrective torques when the foot becomes constrained. Future work will investigate perception-aware foot placement and further sim2real refinement of feedforward torque generation to improve recovery from foot entrapment on discontinuous terrain.

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